RAJA KUMAR

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Summary

Final-year B.Tech CSE student **(CGPA 9.11)** specializing in **AI/ML**, **Generative AI**, and intelligent systems. Experienced in building **RAG** applications, AI agents & vector databases, using Python, FastAPI, Docker, and CI/CD.

Skills

Programming Languages: Java, Python, C

Backend & Frameworks: FastAPI, Flask, Express, REST APIs, SDKs, Backend Development, Next.js, React

AI/ML & Generative AI: Machine Learning, Deep Learning, Generative AI, LLMs, AI Agents, RAG, Prompt Engineering, LangChain, LangGraph, TensorFlow, Evaluation Methods

Cloud, DevOps & Observability: AWS, Docker, Git, GitHub Actions, CI/CD, Observability

Software Engineering: System Design, Software Design, Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Debugging

Responsible AI & Security: Responsible AI, AI Security, Reliability

Projects

Yseek: YouTube Video RAG Engine | Python, Qdrant, faster-whisper, Sentence-Transformers Github

- Processed **200+ hours** of video via **faster-whisper** and **yt-dlp**, engineering a 75s time-window chunker to preserve temporal contexts.
- Built a **Qdrant** retrieval engine indexing **30,000+ vectors**, delivering **250ms** latency and **91% Hit@3** accuracy on multilingual queries.
- Integrated **FastAPI** and YouTube IFrame API to enable precise seek-to-second playback, enhancing user engagement by 25% through interactive video experiences.

AegisRAG: Intelligent Document Retrival | LangGraph, FastAPI, Qdrant, NeMo, FlashRank, Logfire Github

- Architected a production-grade, enterprise-level multi-agent RAG system using **LangGraph** and **FastAPI**, leveraging cyclic intent routing and checkpoint-based memory to resolve **95%** of technical queries.
- Engineered dual-stage retrieval using **Qdrant Cloud**, **3,072-dimensional Gemini Embeddings**, and **FlashRank** cross-encoder reranking to distinguish relevant technical data from noise, achieving **0.99+ semantic relevance** and sub-100ms latency.
- Strengthened AI security using **NVIDIA NeMo Guardrails** and **Portkey LLM Gateway**, implementing input/output safety checks and preventing **96%** of prompt injection attempts.
- Evaluated RAG quality across **60+ documents** using **RAGAS** metrics and **Jaccard similarity**, integrating **Pydantic** validation and **Logfire** tracing for execution monitoring and runtime observability.

OmniAgent: Multi-Agent Trading System | Python, LangGraph, LangChain, Flask Github

- Architected a multi-agent AI trading platform utilizing LangGraph, automating **80%** of routine market analysis and trade orchestrations.
- Integrated GPT-4o, LLaVA, and TA-Lib indicators (RSI, MACD), reducing false trading signals by 30%.
- Constructed high-throughput Flask REST APIs to handle **50+** concurrent agent requests with **<200ms** inference latency.
- Automated cloud infrastructure deployment via **Docker** and **GitHub Actions**, cutting release cycles from days to **under 15 minutes**.

Education

KIIT UNIVERSITY 2023 – 2027

B.Tech in Computer Science and Engineering **CGPA: 9.11**

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Discrete Mathematics, Machine Learning, Data Mining

Work Experience

InAmigos Foundation | *Machine Learning Intern* Aug 2026 – Present

- Developed end-to-end ML pipelines covering **data preprocessing, EDA, and feature engineering** using Python and Scikit-learn, reducing overall data processing time by **30%** across **10,000+** records.
- Built and benchmarked **Linear Regression, Decision Tree, and Random Forest** models using **cross-validation and GridSearchCV**, achieving a **15%** improvement in prediction accuracy and an **R^2 score of 0.92**.

Achievements & Certifications

- Earned **3** professional credentials including AWS Cloud Architecting, AICTE Data Science Internship (Siemens), and Data Science, AI/ML & GenAI, covering **100+ hours** of advanced curriculum.
- Solved **500+** DSA problems and secured **1st Runner-Up** at KIIT Hackathon UiPath Resume Screener.